

Convolution and correlation theorems for the two-dimensional linear canonical transform and its applications

Qiang Feng^{1,2}, Bing-Zhao Li¹ ✉

¹School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 102488, People's Republic of China

²School of Mathematics and Computer Science, Yanan University, Yanan, Shaanxi 716000, People's Republic of China

✉ E-mail: li_bingzhao@bit.edu.cn

ISSN 1751-9675

Received on 22nd January 2015

Revised on 30th June 2015

Accepted on 29th July 2015

doi: 10.1049/iet-spr.2015.0028

www.ietdl.org

Abstract: Convolution and correlation operations are very important in signal processing community, as well as in sampling, filter design and applications. In this study, the authors derive the convolution and correlation theorems for the two-dimensional linear canonical transform (2D LCT). Moreover, they utilise the convolution theorem to investigate the sampling theorem for the band limited signal in the 2D LCT domain. They also discuss multiplicative filter for the band limited signal in the 2D LCT domain which has much lower computational load than the method in the 2D LCT domain.

1 Introduction

One-dimensional (1D) linear canonical transform (LCT) [1–4] has been studied for a quite long time. It has found many applications in optics, radar system analysis, signal processing, filter design, pattern recognition and so on [3–7]. 2D LCT [8–13] which is generalised LCT has more flexible for extra six degree of freedom. Moreover, it is also found important in both signal/image processing communities. Such as filter design, sampling, image encryption, pattern recognition, and many other areas [14–17].

Convolution is a mathematical operation with several applications in pure and applied mathematics such as numerical analysis, numerical linear algebra, and the design and implementation of finite impulse response filters in signal processing [18–20]. Correlation, which is similar to convolution, is another important operation in signal processing, as well as in optics, in pattern recognition, especially in detection applications [21–23]. During the past years, many studies have shown that convolution and correlation operations could be extended to wider domains. For example, convolution and correlation operations for fractional Fourier transform (FRFT) have been proposed [24–27]. Recently, with intensive research of the LCT, convolution and correlation operations have been further extended to the LCT domain [5–7, 28–33], wavelet transform domain [34], and the offset linear canonical transform (OLCT) domain [35].

The 2D convolution and correlation operations are fundamental properties for 2D LCT which are widely used for image processing, optical pattern recognition, sampling, and filter design [17, 27, 36]. On the one hand, up to now, few people have published research results about the 2D convolution and correlation theorem for 2D LCT. On the other hand, as the 2D LCT has found wide applications in signal processing fields, it is theoretically interesting and practically useful to consider the 2D convolution and the 2D correlation theory in the 2D LCT domain.

In this paper, we will discuss convolution and correlation theorems in the 2D LCT domain based on the 2D convolution operation for 2D LCT. The remaining of the paper is organised as follows. Section 2 introduces the definitions and properties of 2D LCT. In Section 3, we obtain the 2D convolution and correlation theorems for 2D LCT in detail. The sampling theorem for 2D LCT and multiplicative filter for the band limited signal in the 2D LCT domain are presented in Section 4. Finally, conclusions are made in Section 5.

2 Preliminaries

The 2D LCT of a function $f(x, y) \in L^2(\mathbb{R}^2)$ can be defined as follows [8–13]

$$F^{A,B}[f(x,y)](u,v) = F_f^{A,B}(u,v) = \begin{cases} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x,y) K_{A,B}(u,v,x,y) dx dy, & b_1 b_2 \neq 0, |A| = |B| = 1 \\ \sqrt{d_1 d_2} e^{i(c_1 d_1 u^2 + c_2 d_2 v^2 / 2)} f(d_1 u, d_2 v), & b_1^2 + b_2^2 = 0 \end{cases} \quad (1)$$

where

$$\begin{aligned} K_{A,B}(u, v, x, y) &= K_A(u, x) K_B(v, y) \\ K_A(u, x) &= C_A e^{i(d_1 u^2 / 2b_1 - ux/b_1 + a_1 x^2 / 2b_2)}, \\ C_A &= \sqrt{\frac{1}{i2\pi b_1}}, \quad A = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \\ K_B(v, y) &= C_B e^{i(d_2 v^2 / 2b_2 - vy/b_2 + a_2 y^2 / 2b_1)}, \\ C_B &= \sqrt{\frac{1}{i2\pi b_2}}, \quad B = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \end{aligned}$$

and A, B are the real matrices.

2D LCT is the generalisation of classical 2D FRFT and the 2D Fourier transform (2D FT). When

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}, \quad \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} = \begin{pmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{pmatrix} \quad (2)$$

the 2D LCT reduces to the 2D FRFT [37–40]. When $\alpha = \beta = \pi/2$ in (2), the 2D LCT reduces to 2D FT [40]. In this paper, the 2D FT is

Table 1 Basic properties of 2D LCT

Basic property	2D LCT
shifting	$F_g^{A,B}(u, v) = e^{(u c_1 \tau_1 + v c_2 \tau_2) - (i/2)(a_1 c_1 \tau_1^2 + a_2 c_2 \tau_2^2)} F^{A,B}(u - a_1 \tau_1, v - a_2 \tau_2)$, where $g(x, y) = f(x - \tau_1, y - \tau_2)$, $\tau_1, \tau_2 \in R$
modulation	$F_g^{A,B}(u, v) e^{(u u_0 d_1 + v v_0 d_2) - (i/2)(u_0^2 b_1 d_1 + v_0^2 b_2 d_2)} F^{A,B}(u - u_0 b_1, v - v_0 b_2)$, $u_0, v_0 \in R$ where $g(x, y) = e^{i(x u_0 + y v_0)} f(x, y)$, $u_0, v_0 \in R$
time-frequency shift	$F_g^{A,B}(u, v) = e^{i(c_1 \tau_1 + d_1 u_0)u + i(c_2 \tau_2 + d_2 v_0)v - i(b_1 c_1 \tau_1 u_0 + b_2 c_2 \tau_2 v_0)} \cdot e^{-(i/2)(a_1 c_1 \tau_1^2 + a_2 c_2 \tau_2^2 + b_1 d_1 u_0^2 + b_2 d_2 v_0^2)} F^{A,B}(u - u_0 b_1 - a_1 \tau_1, v - v_0 b_2 - a_2 \tau_2)$, where $g(x, y) = e^{i(x u_0 + y v_0)} f(x - \tau_1, y - \tau_2)$, $\tau_1, \tau_2, u_0, v_0 \in R$
linearity	$F^{A,B}(\sum_i (a_i f_i(x, y)))(u, v) = a_i \sum_i (F^{A,B} f_i(x, y))(u, v)$, where A, B are the real matrices, and $a_i (i = 1, 2, \dots)$ is real constants
scaling	$F^{A,B}(f(\alpha x, \beta y))(u, v) = \alpha \beta F^{A,B}(f(x, y))(u, v)$, $\alpha, \beta \in R$
multiplication	$F_g^{A,B}(u, v) = (-b_1 \frac{\partial}{\partial u} + i d_1 u) \times (b_2 \frac{\partial}{\partial v} - i d_2 v) F_f^{A,B}(u, v)$ where $g(x, y) = x y f(x, y)$, and $m, n \in \mathbb{N}$
additivity	$F_f^{A_1, B_1}(F_f^{A_2, B_2}(u, v)) = F_f^{A, B}(u, v)$ $A = A_1 A_2$ and $B = B_1 B_2$, $A_i, B_i (i = 1, 2)$ are the real matrices

defined as follows

$$F[f(x, y)](u, v) = F_f(u, v) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) e^{-i(ux+vy)} dx dy \quad (3)$$

and $F_f^{A,B}(u, v)$, $F_f^{\alpha, \beta}(u, v)$, and $F_f(u, v)$ denote 2D LCT, 2D FRFT, and 2D FT of $f(x, y)$, respectively.

The inverse transform of 2D LCT is given by

$$\begin{aligned} f(x, y) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F^{A,B}(u, v) K_{A,B}^*(u, v, x, y) du dv \\ &= C_A^* C_B^* e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F^{A,B}(u, v) \\ &\quad \times e^{i(ux/b_1 + vy/b_2)} e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} du dv \end{aligned} \quad (4)$$

and

$$\begin{aligned} K_{A,B}^*(u, v, x, y) &= K_A^*(u, x) K_B^*(v, y), \\ K_A^*(u, x) &= C_A^* e^{-i(d_1 u^2/2b_1 - ux/b_1 + a_1 x^2/2b_2)} \\ C_A^* &= \sqrt{-\frac{1}{i2\pi b_1}}, \\ K_B^*(v, y) &= C_B^* e^{-i(d_2 v^2/2b_2 - vy/b_2 + a_2 y^2/2b_2)} \\ C_B^* &= \sqrt{-\frac{1}{i2\pi b_2}}, \end{aligned}$$

superscript * denotes complex conjugation, which is also used in the following content of this paper. Moreover, we have

$$K_{A,B}^*(u, v, x, y) = K_{A^{-1}, B^{-1}}(x, y, u, v) = K_{A,B}^{-1}(x, y, u, v)$$

where A^{-1} , B^{-1} are the inverse matrices of A , B , such that

$$A^{-1} = \begin{pmatrix} d_1 & -b_1 \\ -c_1 & a_1 \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} d_2 & -b_2 \\ -c_2 & a_2 \end{pmatrix}$$

The 2D LCT has many properties, some of which are listed in Table 1, for more detailed properties of the 2D LCT, one can refer to [12, 41].

3 Convolution and correlation theorems for 2D LCT

In this section, we derive convolution and correlation theorems for 2D LCT base on the definition and properties of the 2D LCT.

3.1 2D convolution theorems for 2D LCT

The convolution operation for 2D FT is defined

$$f(x, y) * g(x, y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\tau_1, \tau_2) g(x - \tau_1, y - \tau_2) d\tau_1 d\tau_2 \quad (5)$$

then, we find

$$F[f(x, y) \cdot g(x, y)](u, v) = \frac{1}{4\pi^2} F_f(u, v) * F_g(u, v) \quad (6)$$

and

$$F[f(x, y) * g(x, y)](u, v) = F_f(u, v) \cdot F_g(u, v) \quad (7)$$

Based on the expression for the convolution operation for 2D FT, we first define a convolution operation for 2D LCT.

Definition 1: For any function $f(x, y)$, $g(x, y) \in L^2(\mathbb{R}^2)$, the 2D convolution operation for 2D LCT is defined as follows (see (8))

based on Definition 1, firstly, we derive the expressions for the 2D LCT of the product of two functions.

Theorem 1: Let $h(x, y) = f(x, y) \cdot g(x, y)$, $F_h^{A,B}(u, v)$, $F_f^{A,B}(u, v)$, $F_g^{A,B}(u, v)$ denote 2D LCT of the $h(x, y)$, $f(x, y)$ and $g(x, y)$, respectively, then we have (see (9))

where $F_f(u, v)$ denotes 2DFT of $f(x, y)$.

Proof: From the definition of the 2D LCT, we have (see equation at the bottom of the next page)

According to the inverse transform of the 2D LCT, we obtain (see equation at the bottom of the next page)

The proof of Theorem 1 is achieved. $\square$

Next, we derive the expressions for convolution theorem in the 2D LCT domain. This leads to the following convolution theorem:

$$f(x, y) \# g(x, y) = C_A C_B \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\tau_1, \tau_2) g(x - \tau_1, y - \tau_2) e^{-i(a_1 \tau_1 (x - \tau_1)/b_1 + a_2 \tau_2 (y - \tau_2)/b_2)} d\tau_1 d\tau_2 \quad (8)$$

$$F_h^{A,B}(u, v) = \frac{1}{4\pi^2} \frac{1}{|b_1 b_2|} e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \left(\left(F_g^{A,B}(u, v) e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \right) * F_f\left(\frac{u}{b_1}, \frac{v}{b_2}\right) \right) \quad (9)$$

Theorem 2: Let $h(x, y) = f(x, y) \# g(x, y)$ and $F_h^{A,B}(u, v)$, $F_f^{A,B}(u, v)$, $F_g^{A,B}(u, v)$ denote the 2D LCT of the $h(x, y)$, $f(x, y)$ and $g(x, y)$, respectively, then we obtain

$$F_h^{A,B}(u, v) = F_f^{A,B}(u, v) \cdot \tilde{G}(u, v) \quad (10)$$

where

$$\tilde{G}(u, v) = F_g^{A,B}(u, v) e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)}.$$

Proof: From (8), one finds (see (11))

Let $x - s = \tau_1$, $y - t = \tau_2$ in (11), then we have (see (12))

and equivalently, we obtain

$$F_h^{A,B}(u, v) = F_f^{A,B}(u, v) \cdot \tilde{G}(u, v)$$

where

$$\tilde{G}(u, v) = F_g^{A,B}(u, v) e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)}.$$

This completes the proof of Theorem 2. $\square$

Further, the convolution operation in (8) can be expressed as another form (see equation at the bottom of the page)

so, we have (see (13))

see Fig. 1 for realisation of the convolution operation for 2D LCT.

$$F_h^{A,B}(u, v) = C_A C_B e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) g(x, y) e^{-i(ux/b_1 + vy/b_2)} e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} dx dy$$

$$\begin{aligned} F_h^{A,B}(u, v) &= C_A C_B e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) e^{-i(ux/b_1 + vy/b_2)} e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \\ &\quad \times \left(C_A^* C_B^* e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F_g^{A,B}(s, t) e^{i(sx/b_1 + ty/b_2)} e^{-i(d_1 s^2/2b_1 + d_2 t^2/2b_2)} ds dt \right) dx dy \\ &= \frac{1}{4\pi^2} \frac{1}{|b_1 b_2|} e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F_g^{A,B}(s, t) e^{-i(d_1 s^2/2b_1 + d_2 t^2/2b_2)} \\ &\quad \times \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) e^{-i((u-s)x/b_1 + (v-t)y/b_2)} dx dy \right) ds dt \\ &= \frac{1}{4\pi^2} \frac{1}{|b_1 b_2|} e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \left(\left(F_g^{A,B}(u, v) e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \right) * F_f \left(\frac{u}{b_1}, \frac{v}{b_2} \right) \right) \end{aligned}$$

$$\begin{aligned} F_h^{A,B}(u, v) &= C_A C_B \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} e^{-i(ux/b_1 + vy/b_2)} e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \\ &\quad \times \left(C_A C_B \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(s, t) g(x - s, y - t) e^{i(a_1 s((s-x)/b_1 + a_2 t((t-y)/b_2))} ds dt \right) dx dy \end{aligned} \quad (11)$$

$$\begin{aligned} F_h^{A,B}(u, v) &= C_A^2 C_B^2 \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \left(f(s, t) e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} e^{-i(us/b_1 + vt/b_2)} e^{i(a_1 s^2/2b_1 + a_2 t^2/2b_2)} \right) ds dt \\ &\quad \times \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(\tau_1, \tau_2) e^{-i(u\tau_1/b_1 + v\tau_2/b_2)} e^{i(a_1 \tau_1^2/2b_1 + a_2 \tau_2^2/2b_2)} d\tau_1 d\tau_2 \right) \\ &= F_f^{A,B}(u, v) F_g^{A,B}(u, v) e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \end{aligned} \quad (12)$$

$$\begin{aligned} h(x, y) &= f(x, y) \# g(x, y) \\ &= C_A C_B \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\tau_1, \tau_2) g(x - \tau_1, y - \tau_2) e^{-i(a_1 \tau_1((x-\tau_1)/b_1 + a_2 \tau_2(y-\tau_2)/b_2))} d\tau_1 d\tau_2 \\ &= C_A C_B e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\tau_1, \tau_2) e^{i(a_1 \tau_1^2/2b_1 + a_2 \tau_2^2/2b_2)} \\ &\quad \times g(x - \tau_1, y - \tau_2) e^{i(a_1 (x-\tau_1)^2/2b_1 + a_2 (y-\tau_2)^2/2b_2)} d\tau_1 d\tau_2 \\ &= C_A C_B e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \left[\left(f(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right) * \left(g(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right) \right] \end{aligned}$$

$$f(x, y) \# g(x, y) = C_A C_B e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \left[\left(f(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right) * \left(g(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right) \right] \quad (13)$$

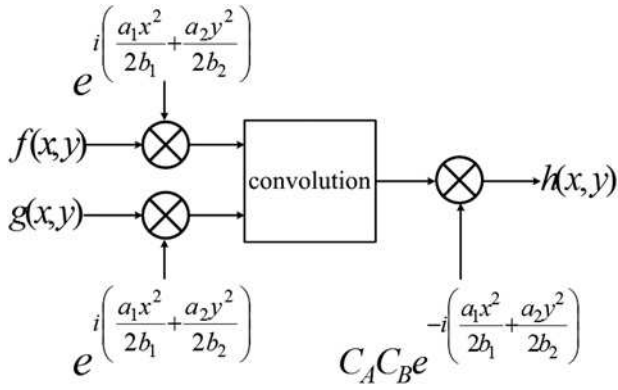


Fig. 1 Convolution operation for 2D LCT

Equation (10) shows the 2D LCT of convolution of two functions is equivalent to simple multiplication of their 2D LCTs. This may be particularly useful in filter design and applications as will be shown later on.

If the parameter matrices $A = (0, 1, -1, 0)$; $B = (0, 1, -1, 0)$ in (9) and (10), then the convolution theorem for 2D LCT reduces to the convolution theorem for 2D FT which is mentioned in (6) and (7).

3.2 2D correlation theorems for 2D LCT

2D correlation, which is similar to 2D convolution, is another important operation in signal processing. Using of the 2D convolution structure for the 2D LCT, we can present a 2D correlation operation in the 2D LCT domain as follows:

Definition 2: For any function $f(x, y), g(x, y) \in L^2(\mathbb{R}^2)$, the 2D correlation operation $\oplus$ for 2D LCT is defined as follows (see (14))

Based on Definition 2, the expressions for the 2D LCT of 2D correlation of two functions can be obtained.

Theorem 3: Let $h(x, y) = f(x, y) \oplus g(x, y)$, $F_h^{A,B}(u, v)$, $F_f^{A,B}(u, v)$, $F_g^{A,B}(u, v)$ denote the 2D LCT of the $h(x, y)$, $f(x, y)$ and $g(x, y)$, respectively, then we have

$$F_h^{A,B}(u, v) = C_A C_B F_f^{A,B}(u, v) \cdot F_g^{A,B}(u, v) e^{i(d_1 u^2 / 2b_1 + d_2 v^2 / 2b_2)} \quad (15)$$

Proof: From the definition of the 2D LCT, we have (see (16))

Let $s - x = \tau_1$, $t - y = \tau_2$ in (16), then we have (see (17))

multiplying both sides of (17) by $e^{-i(d_1 u^2 / 2b_1 + d_2 v^2 / 2b_2)}$, we easily obtain (see equation at the bottom of the page)

and equivalently, we obtain

$$F_h^{A,B}(u, v) = C_A C_B F_f^{A,B}(u, v) \cdot F_g^{A,B}(u, v) e^{i(d_1 u^2 / 2b_1 + d_2 v^2 / 2b_2)}$$

the proof of Theorem 3 is also achieved. $\square$

If the parameter matrices $A = (0, 1, -1, 0)$, $B = (0, 1, -1, 0)$ in (15), then the correlation for 2D FT can be obtained as follows

$$F[f(x, y) \oplus g(x, y)](u, v) = \frac{1}{i2\pi} F_f(u, v) \cdot F_g^*(u, v) \quad (18)$$

4 Applications

Sampling and filtering are two basic theoretical problems in signal processing. In this section, the convolution theorems mentioned above will be utilised to resolve these problems associated with the 2D LCT.

$$f(x, y) \oplus g(x, y) = C_A C_B C_A^* C_B^* \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\tau_1, \tau_2) g^*(\tau_1 - x, \tau_2 - y) e^{i(a_1 x(\tau_1 - x)/b_1 + a_2 y(\tau_2 - y)/b_2)} d\tau_1 d\tau_2 \quad (14)$$

$$F_h^{A,B}(u, v) = F^{A,B}\{f(x, y) \oplus g(x, y)\}(u, v) = C_A C_B \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \left[C_A C_B C_A^* C_B^* \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(s, t) g^*(s - x, t - y) \right. \\ \left. \times e^{i(a_1 x(s - x)/b_1 + a_2 y(t - y)/b_2)} ds dt \right] \left[e^{i(d_1 u^2 / 2b_1 + d_2 v^2 / 2b_2)} e^{-i(ux/b_1 + vy/b_2)} e^{i(a_1 x^2 / 2b_1 + a_2 y^2 / 2b_2)} \right] dx dy \quad (16)$$

$$F_h^{A,B}(u, v) = F^{A,B}\{f(x, y) \oplus g(x, y)\}(u, v) \\ = C_A^2 C_B^2 \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \left(f(s, t) e^{i(d_1 u^2 / 2b_1 + d_2 v^2 / 2b_2)} e^{-i(us/b_1 + vt/b_2)} e^{i(a_1 s^2 / 2b_1 + a_2 t^2 / 2b_2)} \right) ds dt \\ \times \left(C_A^* C_B^* \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g^*(\tau_1, \tau_2) e^{i(u\tau_1/b_1 + v\tau_2/b_2)} e^{-i(a_1 \tau_1^2 / b_1 + a_2 \tau_2^2 / b_2)} \right) d\tau_1 d\tau_2 \\ = C_A C_B F_f^{A,B}(u, v) \left(C_A^* C_B^* \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g^*(\tau_1, \tau_2) e^{i(u\tau_1/b_1 + v\tau_2/b_2)} e^{-i(a_1 \tau_1^2 / 2b_1 + a_2 \tau_2^2 / 2b_2)} \right) d\tau_1 d\tau_2 \quad (17)$$

$$e^{-i(d_1 u^2 / 2b_1 + d_2 v^2 / 2b_2)} F_h^{A,B}(u, v) \\ = C_A C_B F_f^{A,B}(u, v) \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} C_A C_B g(\tau_1, \tau_2) e^{i(d_1 u^2 / 2b_1 + d_2 v^2 / 2b_2)} e^{-i(u\tau_1/b_1 + v\tau_2/b_2)} e^{i(a_1 \tau_1^2 / 2b_1 + a_2 \tau_2^2 / 2b_2)} d\tau_1 d\tau_2 \right)^* \\ = C_A C_B F_f^{A,B}(u, v) \cdot F_g^{A,B}(u, v)$$

4.1 Sampling theory

The sampling process is central in almost any domain, and it explains how to sample continuous signals without aliasing. The sampling theorem expansions for the 2D LCT have been derived in [17], in which the author first derives the spectrum $F_{f_s}^{A,B}(u, v)$ of samples signal $f_s(x, y)$ in the 2D LCT domain, second, apply a low-pass filter to spectrum $F_{f_s}^{A,B}(u, v)$, then, obtain reconstruction formula of original signal $f(x, y)$ which can be written as the following form

$$f(x, y) = F^{A^{-1}, B^{-1}} \left[F_{f_s}^{A,B}(u, v) \cdot H(u, v) \right] (x, y) \quad (19)$$

where $H(u, v)$ is a low-pass filter. It needs two times 2D LCT's calculations.

In this section, using different methods, sampling of band-limited signals in the 2D LCT domain is further investigated. In particular, the formulae of uniform sampling and low-pass reconstruction are obtained [see (25) or (26)]. We can written (25) or (26) as the following form: (see (20))

this equation's form is 2D convolution, and we can use 2D FFT [42] algorithm. In the following section, we will see how to obtain this formula in detail.

Firstly, the uniform sampled signal is defined as follows

$$\begin{aligned} f_s(x, y) &= f(x, y) s_\delta(x, y) \\ &= f(x, y) \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} \delta(x - kT_x) \delta(y - lT_y) \end{aligned} \quad (21)$$

where $f(x, y)$ be an analogue 2D band-limited signal, and T_x and T_y are uniform sampling periods on X - and Y -axes, respectively. $\delta(x - kT_x) \delta(y - lT_y)$ be the uniform impulse train. Its 2D FT is

$$\begin{aligned} F_\delta(u, v) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} \delta(x - kT_x) \delta(y - lT_y) e^{-i(ux+vy)} dx dy \\ &= \frac{4\pi^2}{T_x T_y} \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} \delta\left(u - k \frac{2\pi}{T_x}\right) \delta\left(v - l \frac{2\pi}{T_y}\right) \end{aligned}$$

using the convolution theorem (see Theorem 1), we have (see (22))

From (22), the 2D uniform sampling signal in the 2D LCT domain can be seen as the repetition with a period of $2\pi b_1/T_x$, $2\pi b_2/T_y$, when

$k=0, l=0$, the component of $F_{f_s}^{A,B}(u, v)$ modulates the amplitude of $F_f^{A,B}(u, v)$ by $1/T_x T_y$. When $k \neq 0, l \neq 0$, the component of $F_{f_s}^{A,B}(u, v)$ result in modulation and phase shift of $F_f^{A,B}(u, v)$. If $f(x, y)$ is band limited signal in the 2D LCT domain, then there is no overlap in the $F_{f_s}^{A,B}(u, v)$ after sampling when sampling frequency $\omega_x \geq 2\Omega'_x/b_1$, $\omega_y \geq 2\Omega'_y/b_2$, where $\omega_x = 2\pi/T_x$, $\omega_y = 2\pi/T_y$, while $[-\Omega'_x, \Omega'_x] \times [-\Omega'_y, \Omega'_y]$ is the support interval of $F_f^{A,B}(u, v)$. Thus the $F_f^{A,B}(u, v)$ can be recovered.

Next, we obtain $F_f^{A,B}(u, v)$ through a low-pass filter $\tilde{G}(u, v)$ satisfying (see (23))

where Ω_x, Ω_y be the cutoff frequency. The expression of the above-mentioned signal can be obtained by Theorem 2. From (10), we find

$$\begin{aligned} g(x, y) &= F^{A^{-1}, B^{-1}} \left(\tilde{G}(u, v) e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \right) \\ &= C_A^* C_B^* e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \\ &\quad \times \int_{-\Omega'_x}^{+\Omega'_x} \int_{-\Omega'_y}^{+\Omega'_y} T_x T_y e^{-i(ux/b_1 + vy/b_2)} du dv \\ &= 2 \frac{\sqrt{b_1 b_2}}{i\pi} C_A^* C_B^* T_x T_y e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \\ &\quad \times \frac{\sin(x\Omega_x/b_1) \sin(y\Omega_y/b_2)}{xy} \end{aligned}$$

According to convolution theorem (see Theorem 2) and (22), we have

$$F_{f_s}^{A,B}(u, v) \cdot \tilde{G}(u, v) = F_f^{A,B}(u, v) \quad (24)$$

where $f(x, y) = f_s(x, y)g(x, y)$, from (13), the output of the low-pass filter is as follows: (see equation (25) at the bottom of the next page)

Equation (25) is the reconstruction formula of band-limited signal in the 2D LCT domain. Let $T_x = \pi b_1/\Omega_x$, $T_y = \pi b_2/\Omega_y$ in (25), the interpolation formula (25) can be presented as follows: (see equation (26) at the bottom of the next page)

Equation (26) provides the sampled values of the signal $f(x, y)$. According to the computation complexity, we can see that (26) has the same capability with (19), but less computational complexity than the author's in [17].

When $A = (\cos\theta_1, \sin\theta_1, -\sin\theta_1, \cos\theta_1)$ and $B = (\cos\theta_2, \sin\theta_2, -\sin\theta_2, \cos\theta_2)$ or $A = (0, 1, -1, 0)$ and $B = (0, 1, -1, 0)$, then we

$$f(x, y) = f_s(x, y) \# g(x, y) = C_A C_B e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \left(f_s(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right) * \left(g(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right) \quad (20)$$

$$\begin{aligned} F_{f_s}^{A,B}(u, v) &= \frac{1}{4\pi^2 b_1 b_2} e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \left(\left(F_f^{A,B}(u, v) e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \right) * \left(F_\delta\left(\frac{u}{b_1}, \frac{v}{b_2}\right) \right) \right) \\ &= \frac{1}{T_x T_y} e^{i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \left(\left(F_f^{A,B}(u, v) e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \right) * \left(\sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} \delta\left(u - k \frac{2\pi}{T_x} b_1\right) \delta\left(v - l \frac{2\pi}{T_y} b_2\right) \right) \right) \end{aligned} \quad (22)$$

$$\tilde{G}(u, v) = \begin{cases} T_x T_y, & |u| \leq \Omega_x, |v| \leq \Omega_y \\ 0, & \text{others} \end{cases}, \quad \Omega_x \in [-\Omega'_x, b_1 \omega_x - \Omega'_x], \quad \Omega'_y \in [-\Omega'_y, b_2 \omega'_y - \Omega'_y] \quad (23)$$

$$\begin{aligned} f(x, y) &= f_s(x, y) g(x, y) \\ &= C_A C_B e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \left(f_s(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right) * \left(g(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right) \\ &= e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} f(kT_x, lT_y) e^{i(a_1 (kT_x)^2/2b_1 + a_2 (lT_y)^2/2b_2)} \frac{T_x \sin((x - kT_x)\Omega_x/b_1)}{\pi(x - kT_x)} \frac{T_y \sin((y - lT_y)\Omega_y/b_2)}{\pi(y - lT_y)} \end{aligned} \quad (25)$$

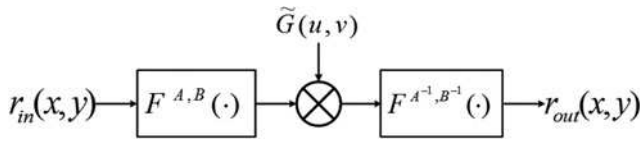


Fig. 2 Multiplicative filter in the 2D LCT domain

can obtain the reconstruction formula of 2D FRFT [17] or reconstruction formula 2D FT [17], respectively.

4.2 Multiplicative filter in the 2D LCT domain

Many papers [5–7, 15, 21, 27–29, 31] discussed the use of the multiplicative filter designed by the FRFT or LCT to remove noise or distortion. Here, we generalise the previous works to the 2D LCT domain. Based on Theorem 2, we can discuss the design methods and the performance of the multiplicative filter designed by the 2D LCT. The model of the multiplicative filter in the 2D LCT domain is shown in Fig. 2. The effect of the multiplicative filter in Fig. 2 can be written in the following equation

$$r_{\text{out}}(x, y) = F^{A^{-1}, B^{-1}} \left[F_{r_{\text{in}}}^{A, B}(u, v) \cdot \tilde{G}(u, v) \right] (x, y) \quad (27)$$

where $F_{r_{\text{in}}}^{A, B}(u, v)$ denotes the 2D LCT of received signal $r_{\text{in}}(x, y)$. Many models of multiplicative filter can be achieved by different transform functions such as low pass, high pass, band pass and band stop and so on. For example, we are interested only in the frequency spectrum of the 2D LCT in the region $[u_x^1, u_x^2] \times [u_y^1, u_y^2]$ of the designed signal $f(x, y)$. We also suppose that the 2D LCT parts of the received signal $r_{\text{in}}(x, y)$, $F_{r_{\text{in}}}^{A, B}(u, v)$, and $F_n^{A, B}(u, v)$, have no overlapping or minimal overlapping. According to Theorem 2, we choose function $\tilde{G}(u, v)$ in (10) as the transfer function of the multiplicative filter. However, $\tilde{G}(u, v)$ in Fig. 2 can be expressed as follows

$$\tilde{G}(u, v) = F_g^{A, B}(u, v) e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} \quad (28)$$

The transfer function $\tilde{G}(u, v)$ can be designed as a constant over $[u_x^1, u_x^2] \times [u_y^1, u_y^2]$ and zero or of rapid decay outside that region.

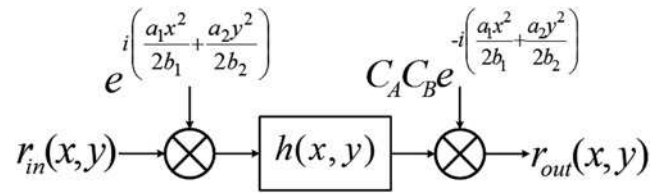


Fig. 3 Method to achieve the multiplicative filters in the 2D LCT domain by convolution in the time domain

Passing the output of the filter yields that part of the spectrum of $f(x, y)$ over $[u_x^1, u_x^2] \times [u_y^1, u_y^2]$. By reversibility property of 2D LCT, the designed signal $f(x, y)$ can be obtained. The above-mentioned multiplicative filter in the 2D LCT domain can also be achieved through convolution in the time domain. Fig. 3 shows a method of realising the multiplicative filter in the 2D LCT domain by convolution in the time domain. From Fig. 3, the output of the filter is expressed as (see (29))

where the convolution function $h(x, y)$ in Fig. 3 can be designed as

$$h(x, y) = g(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \quad (30)$$

and (see (31))

where $\tilde{g}(x, y)$ is the inverse 2D FT of $\tilde{G}(u, v)$. Therefore, we obtain

$$h(x, y) = \frac{1}{i\sqrt{b_1 b_2}} \tilde{g}\left(\frac{x}{b_1}, \frac{y}{b_2}\right) \quad (32)$$

substituting (32) into (29), one finds (see (33))

Next, we will prove that the output of filter shown in Fig. 3 is same as that in Fig. 2. Substituting (30) into (29), we obtain (see (34)) in fact, (34) is equivalent to (33), and according to Theorem 2, we

$$f(x, y) = e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \sum_{k=-\infty}^{+\infty} \sum_{l=-\infty}^{+\infty} f(kT_x, lT_y) \times e^{i(a_1 (kT_x)^2/2b_1 + a_2 (lT_y)^2/2b_2)} \frac{\sin((x - kT_x)\Omega_x/b_1)}{(x - kT_x)\Omega_x/b_1} \frac{\sin((y - lT_y)\Omega_y/b_2)}{(y - lT_y)\Omega_y/b_2} \quad (26)$$

$$r_{\text{out}}(x, y) = C_A C_B e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \left[r_{\text{in}}(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right] * h(x, y) \quad (29)$$

$$\begin{aligned} g(x, y) &= F^{A^{-1}, B^{-1}} \left[F_g^{A, B}(u, v) \right] (x, y) \\ &= C_A^* C_B^* e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F_g^{A, B}(u, v) e^{i(ux/b_1 + vy/b_2)} e^{-i(d_1 u^2/2b_1 + d_2 v^2/2b_2)} du dv \\ &= C_A^* C_B^* e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{i(ux/b_1 + vy/b_2)} \tilde{G}(u, v) du dv \\ &= \frac{1}{i\sqrt{b_1 b_2}} e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \tilde{g}\left(\frac{x}{b_1}, \frac{y}{b_2}\right) \end{aligned} \quad (31)$$

$$r_{\text{out}}(x, y) = C_A C_B e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \left[r_{\text{in}}(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right] * \left[\frac{1}{i\sqrt{b_1 b_2}} \tilde{g}\left(\frac{x}{b_1}, \frac{y}{b_2}\right) \right] \quad (33)$$

$$r_{\text{out}}(x, y) = C_A C_B e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \left[r_{\text{in}}(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right] * \left[g(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} \right] \quad (34)$$

have

$$F_{r_{\text{out}}}^{A,B}(u, v) = F_{r_{\text{in}}}^{A,B}(u, v) \cdot \tilde{G}(u, v) \quad (35)$$

then, we achieve the proof by using formulae (27), (32), and (34).

In an actual application, the method shown in Fig. 3 has better performance than that in Fig. 2 for less computation complexity. The reason is that the major computational load of the former method is convolution and the calculation of $g(x, y)$, but these two steps can be done by the 2D FFT algorithm [42]. While the latter methods requirement is of calculating 2D LCT twice, and there exists still no satisfactory fast digital algorithm for 2D LCT so far. Therefore, the former method has the same capability, but less computational complexity, for the avoidance of cumbersome calculation for 2D LCT.

Let us analyse the computation complexity of the multiplicative filter achieved in the time domain from (33) in detail. From the 2D FT, we rewrite (33) as follows: (see (36))

where

$$\varphi_1(x, y) = r_{\text{in}}(x, y) e^{i(a_1 x^2/2b_1 + a_2 y^2/2b_2)}, \quad \varphi_2(x, y) = \tilde{g}(x/b_1, y/b_2),$$

from inverse of 2D FT, we obtain (see (37))

where $\varphi_3(x, y) = F_{\varphi_1}(u, v) \cdot \tilde{G}(b_1 u, b_2 v)$. From (37), the 2D FFT can be used to reduce the computation complexity of this filtering method. For N point of samples, the computation complexity of 2D FFT is $O(N^2(\log_2^N))$, so the computational complexity of the former method for N point of samples is $O(N^2(\log_2^N))$, but for the latter is $O(N^4)$.

5 Conclusions

In this paper, first, we introduce the definition and some properties of 2D LCT. Second, we present the 2D convolution and correlation for 2D LCT based on the 2D convolution for 2D FT. Then, convolution theorems for 2D LCT have been derived in detail, presenting the convolution theorem in the 2D LCT domain as well as a product between two functions in the time domain. We also investigate the expressions for 2D LCT of correlation operation of two functions based on Theorem 2, and obtain the correlation theorems for 2D LCT. Finally, as an application, utilising convolution theorems derived in this paper, sampling of band-limited signals in the 2D LCT domain has been further investigated. The formulas of uniform sampling and low-pass reconstruction are obtained. Moreover, the multiplicative filter in the 2D LCT domain has also been discussed in this paper. A practical method to achieve the multiplicative filter through convolution in the time domain is proposed. This method can be realised by classical 2D FFT and has the same capability, but less computational complexity compared with the method achieved in the 2D LCT domain.

6 Acknowledgments

The authors thank the editor and the anonymous referees for their valuable comments and suggestions that improved the clarity and

quality of this manuscript. This work was supported in part by the National Natural Science Foundation of China (no. 61171195), and also supported by Program for New Century Excellent Talents in University (no. NCET-12-0042).

7 References

- Moshinsky, M., Quesne, C.: 'Linear canonical transformations and their unitary representations', *J. Math. Phys.*, 1971, **12**, pp. 1772–1783
- Wolf, K.B.: 'Construction and properties of canonical transforms, in Integral Transform in Science and Engineering' (Plenum, New York, 1979)
- Ozaktas, H.M., Zalevsky, Z., Kutay, M.A.: 'The fractional Fourier transform with applications in optics and signal processing' (Wiley, New York, 2001)
- Pei, S.C., Ding, J.J.: 'Eigenfunctions of linear canonical transform', *IEEE Trans. Signal Process.*, 2002, **50**, (1), pp. 11–26
- Deng, B., Tao, R., Wang, Y.: 'Convolution theorems for the linear canonical transform and their applications', *Sci. Chin. Ser. F, Inf. Sci.*, 2006, **49**, (5), pp. 592–603
- Wei, D.Y., Ran, Q.W., Li, Y.M., *et al.*: 'A convolution and product theorem for the linear canonical transform', *IEEE Signal Process. Lett.*, 2009, **16**, (10), pp. 853–856
- Wei, D.Y., Ran, Q.W., Li, Y.M.: 'A convolution and correlation theorem for the linear canonical transform and its application', *Circuits Syst. Signal Process.*, 2012, **31**, (1), pp. 301–312
- Xu, G.L., Wang, X.T., Xu, X.B.: 'Generalized Hilbert transform and its properties in 2D LCT domain', *Signal Process.*, 2009, **89**, (7), pp. 1395–1402
- Wolf, K.B.: 'Integral transforms in science and engineering' (Plenum Press, New York, America, 1979)
- Ozaktas, H.M., Koc, A., Sari, I., *et al.*: 'Efficient computation of quadratic-phase integrals in optics', *Opt. Lett.*, 2006, **31**, (1), pp. 35–37
- Healy, J.J., Sheridan, J.T.: 'Fast linear canonical transforms', *J. Opt. Soc. Am. A*, 2010, **27**, (1), pp. 21–30
- Alieva, T., Bastiaans, M.J.: 'Properties of the linear canonical integral transformation', *J. Opt. Soc. Am. A*, 2007, **24**, (11), pp. 3658–3665
- Oktem, F.S., Ozaktas, H.M.: 'Equivalence of linear canonical transform domains to fractional Fourier domains and the bicanonical width product: a generalization of the space-bandwidth product', *J. Opt. Soc. Am. A*, 2010, **27**, (8), pp. 1885–1895
- Sahin, A., Kutay, M.A., Ozaktas, H.M.: 'Nonseparable two-dimensional fractional Fourier transform', *Appl. Opt.*, 1998, **37**, (23), pp. 5444–5453
- Kutay, M.A., Ozaktas, H.M.: 'Optimal image restoration with the fractional Fourier transform', *J. Opt. Soc. Am. A*, 1998, **15**, (4), pp. 825–833
- Pei, S.C., Ding, J.J.: 'Two-dimensional affine generalized fractional Fourier transform', *IEEE Trans. Signal Process.*, 2001, **49**, (4), pp. 878–897
- Li, B.Z., Xu, T.Z.: 'Spectral analysis of sampled signals in the linear canonical transform domain', *Math. Probl. Eng.*, 2012, **2012**, pp. 1–19, doi: 10.1155/2012/536464
- Ebling, J., Scheuermann, G.: 'Clifford Fourier transform on vector fields', *IEEE Trans. Vis. Comput. Graphics*, 2005, **11**, (4), pp. 469–479
- Hitzer, E., Mawardi, B.: 'Clifford Fourier transform on multivector fields and uncertainty principles for dimensions', *Adv. Appl. Clifford Algebras*, 2008, **18**, (3–4), pp. 715–736
- Mawardi, B., Hitzer, E.: 'Clifford Fourier transformation and uncertainty principle for the Clifford geometric algebra', *Adv. Appl. Clifford Algebras*, 2006, **16**, (1), pp. 41–61
- Pei, S.C., Ding, J.J.: 'Relations between fractional operations and time-frequency distributions and their applications', *IEEE Trans. Signal Process.*, 2001, **49**, (8), pp. 1638–1655
- Akay, O., Boudreaux-Bartels, G.F.: 'Fractional convolution and correlation via operator methods and application to detection of linear FM signals', *IEEE Trans. Signal Process.*, 2001, **49**, (5), pp. 979–993
- Torres, R., Pellat, F.P., Torres, Y.: 'Fractional convolution, fractional correlation and their translation invariance properties', *Signal Process.*, 2010, **90**, (6), pp. 1976–1984
- Almeida, L.B.: 'Product and convolution theorems for the fractional Fourier transform', *IEEE Signal Process. Lett.*, 1997, **4**, (1), pp. 15–17
- Zayed, A.I.: 'A product and convolution theorems for the fractional Fourier transform', *IEEE Signal Process. Lett.*, 1998, **5**, (4), pp. 101–103
- Singh, A.K., Saxena, R.: 'On convolution and product theorems for Frft', *Wirel. Pers. Commun.*, 2012, **65**, (1), pp. 189–201
- Singh, A.K., Saxena, R.: 'Recent developments in FRFT, DFRFT with their applications in signal and image processing', *Recent Patents Eng.*, 2011, **5**, (2), pp. 113–138
- Wei, D.Y., Ran, Q.W., Li, Y., *et al.*: 'New convolution theorem for the linear canonical transform and its translation invariance property', *Opt. Int. J. Light Electron. Opt.*, 2012, **123**, (16), pp. 1478–1481

$$r_{\text{out}}(x, y) = -\frac{1}{2\pi b_1 b_2} e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} F^{-1} \left[F_{\varphi_1}(u, v) \cdot F_{\varphi_2}(u, v) \right] \quad (36)$$

$$\begin{aligned} r_{\text{out}}(x, y) &= -\frac{1}{2\pi b_1 b_2} e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} F^{-1} \left[F_{\varphi_1}(u, v) \cdot b_1 b_2 \tilde{G}(b_1 u, b_2 v) \right] \\ &= -\frac{1}{2\pi} e^{-i(a_1 x^2/2b_1 + a_2 y^2/2b_2)} F^{-1} \left[F_{\varphi_3}(u, v) \right] \end{aligned} \quad (37)$$

- 29 Shi, J., Liu, X.P., Zhang, N.T.: 'Generalized convolution and product theorems associated with linear canonical transform', *Signal Image Video Process.*, 2014, **8**, (5), pp. 967–974
- 30 Navdeep, G., Kulbir, S.: 'A modified convolution and product theorem for the linear canonical transform in quantum mechanics', *Int. J. Appl. Math. Comput. Sci.*, 2013, **23**, (3), pp. 685–695
- 31 Zulfajar, M.B., Ashino, R.: 'Convolution and correlation theorem for linear canonical transform and properties', *Information*, 2014, **17**, (6), pp. 2509–2521
- 32 Navdeep, G., Kulbir, S.: 'Modified correlation theorem for the linear canonical transform with representation transformation in quantum mechanics', *Signal Image Video Process.*, 2014, **8**, (3), pp. 595–601
- 33 Guo, L.Q., Zhu, M., Ge, X.H.: 'Reduced biquaternion canonical transform, convolution and correlation', *Signal Process.*, 2011, **91**, (8), pp. 2147–2153
- 34 Wei, D.Y., Li, Y.M.: 'Generalized wavelet transform based on the convolution operator in the linear canonical transform domain', *Opt. Int. J. Light Electron. Opt.*, 2014, **125**, (16), pp. 4491–4496
- 35 Xiang, Q., Qin, K.Y.: 'Convolution, correlation, and sampling theorems for the offset linear canonical transform', *Signal Image Video Process.*, 2014, **8**, (3), pp. 433–442
- 36 Boussakta, S., Holt, A.G.J.: 'New two dimensional transforms', *Electron. Lett.*, 1993, **29**, (11), pp. 949–950
- 37 Ozaktas, H.M., Kutay, M.A., Zalevsky, Z.: 'The fractional Fourier Transform with applications in optics and signal processing' (John Wiley, New York, 2001)
- 38 Namias, V.: 'The fractional order Fourier transform and its application to quantum mechanics', *J. Inst. Math. Appl.*, 1980, **25**, (3), pp. 241–265
- 39 Ozaktas, H.M., Mendlovic, D.: 'Fractional Fourier optics', *J. Opt. Soc. Am. A*, 1995, **12**, (4), pp. 743–751
- 40 Ding, J.J., Pei, S.C.: 'Eigenfunctions and self-imaging phenomena of the two-dimensional non separable linear canonical transform', *J. Opt. Soc. Am. A*, 2011, **28**, (2), pp. 82–95
- 41 Xu, T.Z., Li, B.Z.: 'Linear canonical transform and its applications' (Science Press, Beijing, China, 2013)
- 42 Wang, M., Lee, E.B.: '2-D FFT algorithm by matrix factorization in a 2-D space', *Multidim. Syst. Signal Process.*, 1994, **5**, (1), pp. 61–74